

Validation of an optimized tissue-agnostic genome-wide methylome enrichment assay to predict clinical outcomes in patients treated with pembrolizumab.

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Background: Recent work from the INSPIRE study (PMID38393391) suggests that kinetics of cell-free DNA (cfDNA) methylation profiles reflect immunotherapy treatment response in solid tumors. Here we provide validation data of a tissue-agnostic, genome-wide methylation enrichment assay based on cell free methylated DNA immunoprecipitation and high throughput sequencing (cfMeDIP-seq) designed for clinical use, to determine response to immunotherapy. **Methods:** This study utilizes samples and clinical data from the INSPIRE study, a single-institution investigator-initiated phase II study of pembrolizumab in multiple solid tumors given every 3 weeks (NCT02644369). A prior published analysis of cfMeDIP used TCGA to develop a classifier and demonstrated an association of response to immunotherapy. In contrast, in this analysis, **a novel quantitative and highly specific measurement of ctDNA was estimated using a generative machine learning model trained on differentially methylated regions identified from a large cfMeDIP methylome atlas from individuals with and without cancer.** In a blinded validation analysis, Firth's logistic regressions were used to test differences in objective response (ORR) and clinical benefit rate (CBR) defined as complete or partial response or stable disease $> / = 6$ cycles between patients with a **decrease in ctDNA from baseline to cycle 3 of treatment**, and those with an increase in ctDNA. Sensitivity for no objective response, specificity for objective response, and positive and negative predictive values (PPV and NPV) were summarized. **Cox regressions and log-rank tests were used to evaluate differences in progression-free survival (PFS) and overall survival (OS) between the two groups.** **Results:** The analysis included 64 unique patients with a median follow up of 18.43 months (a total of 128 samples), including head & neck (n = 9), triple negative breast (n = 10), ovarian (n = 11), melanoma (n = 7), and other mixed solid tumor types (n = 27). A decrease in ctDNA was associated with significantly better objective response than an increase [odds ratio (OR) 33.89 (4.07, 44426.47), $p = 0.0001$], 58% sensitivity, 100% specificity, 100% PPV and 35% NPV. Significantly better CBR [OR 10.17 (2.74, 55.74), $p = 0.0002$] was also observed. A decrease in ctDNA was associated with significantly better PFS [hazard ratios (HR) 0.28 (0.15, 0.49) $p < 0.0001$] and OS [HR 0.42 (0.24, 0.76) $p < 0.003$]. **Conclusions:** A clinical tissue-agnostic, genome-wide methylome enrichment approach using cfMeDIP-seq accurately predicts clinical outcomes in patients treated with pembrolizumab in multiple advanced solid tumors. This test provides **relative quantification of methylated ctDNA to predict response to immunotherapy** and does not require tumor tissue. This analysis highlights potential generalizability across tumor types in response monitoring. Clinical trial information: NCT02644369. Research Sponsor: Adela, Inc; Merck.